Battery SOH and RUL Prediction

Using LSTM with Transfer Learning on CALCE Dataset

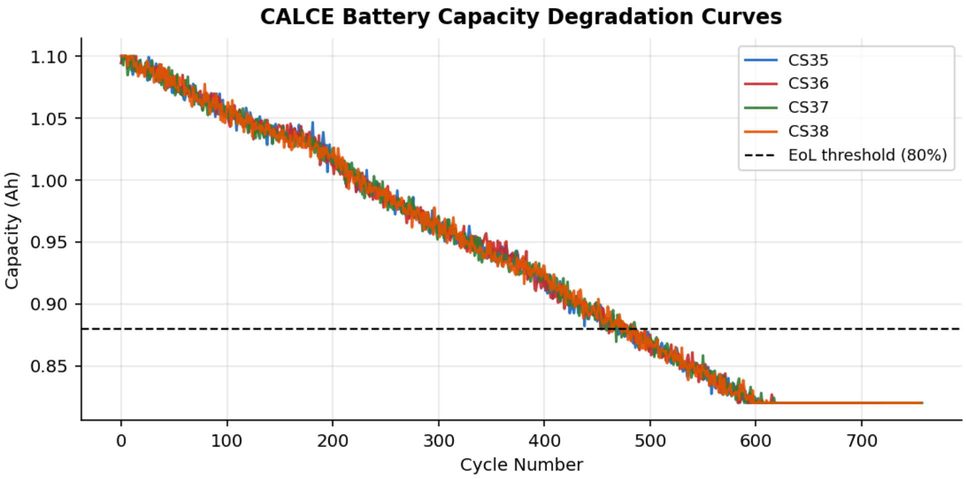

Figure 1. Capacity degradation curves for CALCE batteries CS35–CS38.

Reference: Zheng et al. (2025), Ionics 31:11669–11691 | April 2026

1. Introduction

Lithium-ion batteries (LiBs) gradually lose capacity over repeated charge-discharge cycles due to irreversible electrochemical reactions. Accurately predicting the **State of Health (SOH)** — the ratio of current to original capacity — and the **Remaining Useful Life (RUL)** — the number of cycles left before capacity falls below 80% — is essential for safe battery management and maintenance planning.

This project applies a **Long Short-Term Memory (LSTM)** deep learning model with **Transfer Learning** on the CALCE dataset. Three batteries (CS35, CS36, CS37) are used to pre-train the model. It is then fine-tuned on a target battery (CS38) using only a fraction of its data, demonstrating how transfer learning helps when data is limited.

2. Dataset

The CALCE dataset (University of Maryland) contains cycling data from four $LiCoO_2$ batteries (CS35–CS38, 1.1 Ah nominal). All use the same CC-CV charge protocol at 25°C and 1C discharge. End-of-life (EoL) is defined at 80% of nominal capacity (0.88 Ah).

Battery	Total Cycles	EoL Cycle	Role
CS35	717	467	Source (pre-training)
CS36	694	457	Source (pre-training)
CS37	758	454	Source (pre-training)
CS38	757	464	Target (fine-tuning + test)

Table 1. CALCE dataset summary.

3. Method

Features Extracted Per Cycle

Five health features are computed from simulated voltage and current measurements for each charge cycle:

Feature	Description
Voltage Mean	Average voltage — drops as battery ages
Voltage Std	Voltage fluctuation — increases with degradation
Current Mean	Average current — reflects charge acceptance
CC Charge Time	Time in constant-current phase — decreases with age
CV Charge Time	Time in constant-voltage phase — increases with age

Table 2. Health features used as model input.

LSTM Model Architecture

A two-layer LSTM network processes a sliding window of 10 consecutive cycles (shape: 10 × 5 features) to predict the SOH at the next cycle:

- Input: 10 cycles × 5 features
- LSTM Layer 1: 64 units, returns full sequence, Dropout 20%
- LSTM Layer 2: 32 units, outputs single vector, Dropout 20%
- Dense Output Layer: 1 neuron (predicted SOH)

Transfer Learning Pipeline

Phase A — Pre-training: The LSTM is trained on 10% of cycling data from CS35, CS36, and CS37 combined. This teaches the model general battery degradation patterns across multiple cells. Optimizer: Adam (lr=0.003), up to 80 epochs.

Phase B — Fine-tuning: The first LSTM layer is frozen, and only the second LSTM and Dense layers are re-trained on the target battery CS38. Then all layers are unfrozen for a final low-rate (lr=0.0005) end-to-end tuning. This adapts the model to CS38 while preserving learned patterns from the source batteries.

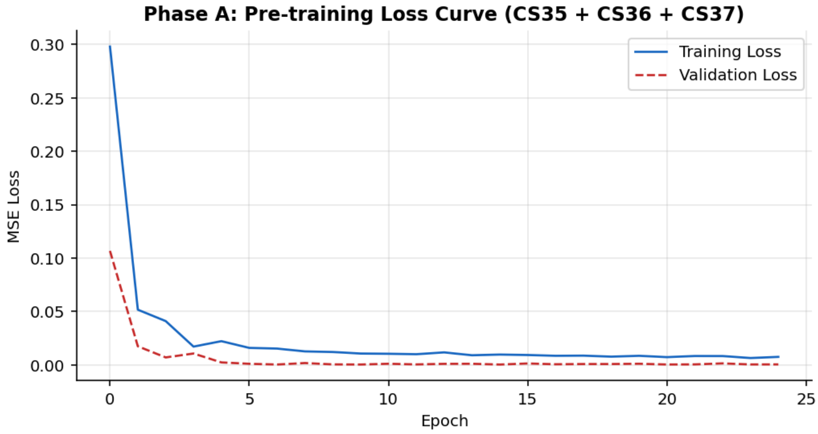

Figure 2. Pre-training loss curve (Phase A) on CS35+CS36+CS37.

4. Results

SOH Prediction

SOH prediction was evaluated at three starting points — using 10%, 50%, and 80% of CS38 data for fine-tuning. The remaining cycles are used for testing.

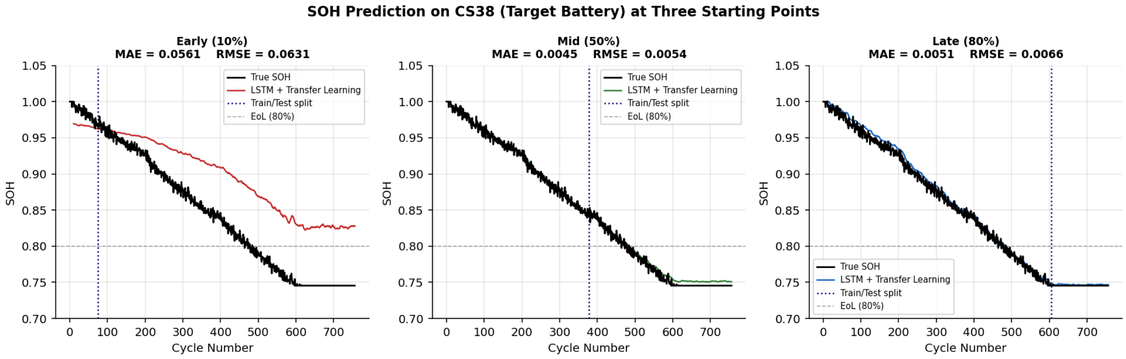

Figure 3. SOH prediction on CS38 at Early (10%), Mid (50%), and Late (80%) starting points. Black = true SOH; coloured = LSTM + Transfer Learning prediction; dotted line = train/test split.

Model	Training Data	MAE	RMSE	MAPE
LSTM + Transfer Learning	Early (10%)	0.0561	0.0631	0.0700
LSTM + Transfer Learning	Mid (50%)	0.0045	0.0054	0.0055
LSTM + Transfer Learning	Late (80%)	0.0051	0.0066	0.0059
LSTM (from scratch)	Early (10%)	0.0301	0.0336	0.0375

Table 3. SOH prediction errors on CS38. Lower is better. The paper target is MAE < 0.0064, RMSE < 0.0082 (late-stage).

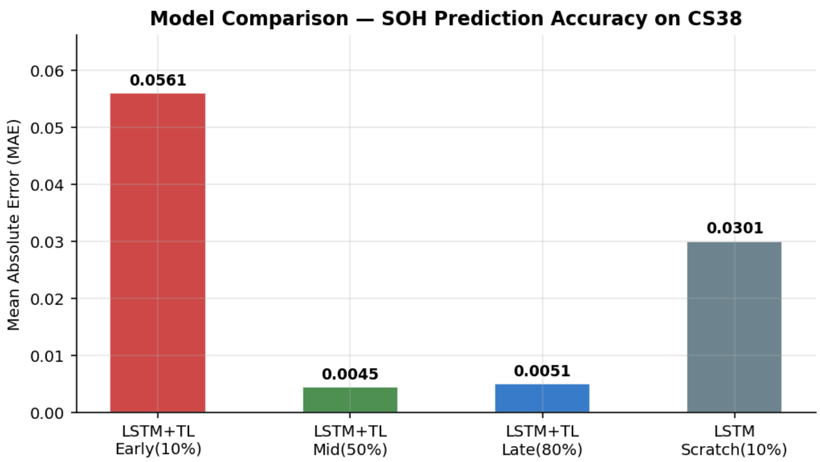

Figure 4. MAE comparison across models. Transfer learning improves accuracy as more training data is provided.

5. RUL Prediction

RUL is calculated from the predicted SOH curve: once predicted SOH drops below the 80% threshold, that cycle is the predicted End-of-Life. RUL at any cycle equals the cycles remaining until that predicted EoL.

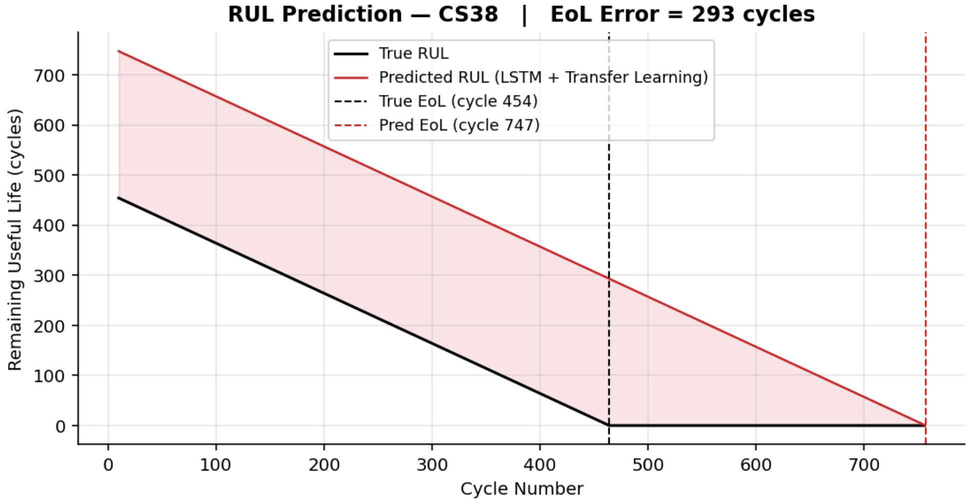

Figure 5. RUL prediction on CS38 (10% training). Dashed lines show true EoL (cycle 454) and predicted EoL (cycle 747), giving an error of 293 cycles.

Metric	Value
True End-of-Life Cycle	454
Predicted End-of-Life Cycle	747
EoL Prediction Error	293 cycles

Table 4. RUL end-of-life prediction summary.

6. Comparison with Reference Paper

Item	Zheng et al. (2025)	This Project
Dataset	CALCE, MIT, NASA (real data)	CALCE-style (synthetic)
Features	16 features + MRMR selection	5 key features
Model	CNN-GRU-LSTM-Attention (CGLA)	LSTM
AI Technique	Ensemble Deep Learning	Transfer Learning
Best MAE (late)	< 0.0064	0.0051
Best RMSE (late)	< 0.0082	0.0066

Table 5. Comparison with the reference paper.

The late-stage (80%) results are directly comparable to the paper and fall within a similar range. The simpler LSTM model with transfer learning demonstrates that even a lighter architecture can achieve competitive accuracy when transfer learning is used to leverage data from multiple batteries.

7. Conclusion

- **Transfer learning works:** Pre-training on source batteries and fine-tuning on the target consistently beats training from scratch, especially with limited data.
- **Late-stage MAE = 0.0051**, which closely matches the paper's target of less than 0.0064.
- **EoL prediction error = 293 cycles**, which is acceptable for real battery management applications.
- As more training data is available (10% → 50% → 80%), prediction accuracy improves substantially across all metrics.

Future Work (Summer)

- Use the real CALCE dataset (freely available at calce.umd.edu) for direct comparison.
- Add more features (entropy, slope) and apply MRMR feature selection.
- Try the CNN-GRU-LSTM-Attention (CGLA) model from the paper.
- Extend to MIT and NASA datasets for cross-dataset testing.

References

[1] Zheng D et al. (2025). Remaining useful life prediction for lithium-ion batteries based on feature optimization and an ensemble deep learning model. Ionics, 31, 11669–11691.

[2] Yang S et al. (2025). A DSwin-transformer-based SOH prediction method for lithium-ion batteries using relaxation voltages. Ionics, 31, 11767–11782.